COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 12 : SLO 2 — Tutorial: IEEE 754 Double Precision Encoding
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Section C: Activity — Encode -25.625 in IEEE 754 Double Precision

• Convert the magnitude: 25 = 11001, 0.625 = 0.101 → 11001.101
• Normalize: 1.1001101 × 2^4
• Sign bit: 1 (negative)
• Biased exponent: 4 + 1023 = 1027 = 10000000011 (11-bit)
• Mantissa (52-bit): 1001101000000000000000000000000000000000000000000000
• IEEE 754 Double Precision: 1 10000000011
1001101000000000000000000000000000000000000000000000
• (Hex: C039 A000 0000 0000)

Section D: Role-Based Play (Analyze)

• Roles: Decimal Converter, Normalizer, Exponent Encoder (bias 1023, 11-bit), Sign Bit Decider, Mantissa Extractor (52-bit fraction), Encoder Assembler (final 64-bit word).
• Execution: each student completes their stage and passes the result on; the final student reads out the full 64-bit binary.
• Discussion: if the number is too large for the exponent range it overflows to infinity; if too small it underflows toward denormalized numbers or zero — precision is limited by the 52-bit mantissa.